

PBOS: Pneumatic Beverage Opening System – Mechanical Design, Pneumatic Actuation, and Prototype Verification

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Abstract

This technical report presents the engineering design, pneumatic actuation architecture, and functional validation of the **Pneumatic Beverage Opening System (PBOS)**. The project addresses the automation and mechanical standardization of beverage cap removal by integrating fluid power actuators into an ergonomic benchtop mechanism. The system was modeled via parametric CAD software to optimize mechanical leverage, structural stability, and actuator alignment. Following dimensional verification, a physical prototype was constructed using precision-machined elements and pneumatic control components. Experimental testing demonstrated reliable cap detachment cycles with minimal operator intervention. This paper details the structural layout, kinematic principles, pneumatic circuit architecture, and experimental findings of the PBOS prototype.

Keywords: Pneumatics, Mechanical Automation, Beverage Cap Decapper, CAD/CAM Prototyping, Actuation Systems, Applied Mechanics.

1. Introduction

Manual bottle and beverage opening operations, while fundamentally simple, present challenges regarding operational fatigue, cycle repeatability, and process standardization in commercial or service environments. The **Pneumatic Beverage Opening System (PBOS)** was developed to transform a conventional manual action into a controlled, repeatable, and semi-automated mechanical process.

The primary objectives of the PBOS project were:

1. To design a compact, reliable mechanical apparatus utilizing compressed air for actuation.
2. To achieve a secure grip and mechanical advantage capable of detaching standard crown/twist-off beverage caps without structural failure or liquid spillage.
3. To document the rapid prototyping cycle from parametric CAD design to physical fabrication.

2. System Architecture & Mechanical Design

2.1 Mechanical Frame and Kinematics

The structural skeleton of PBOS was designed to withstand the axial and reaction forces generated during cap disengagement. Key mechanical parameters include:

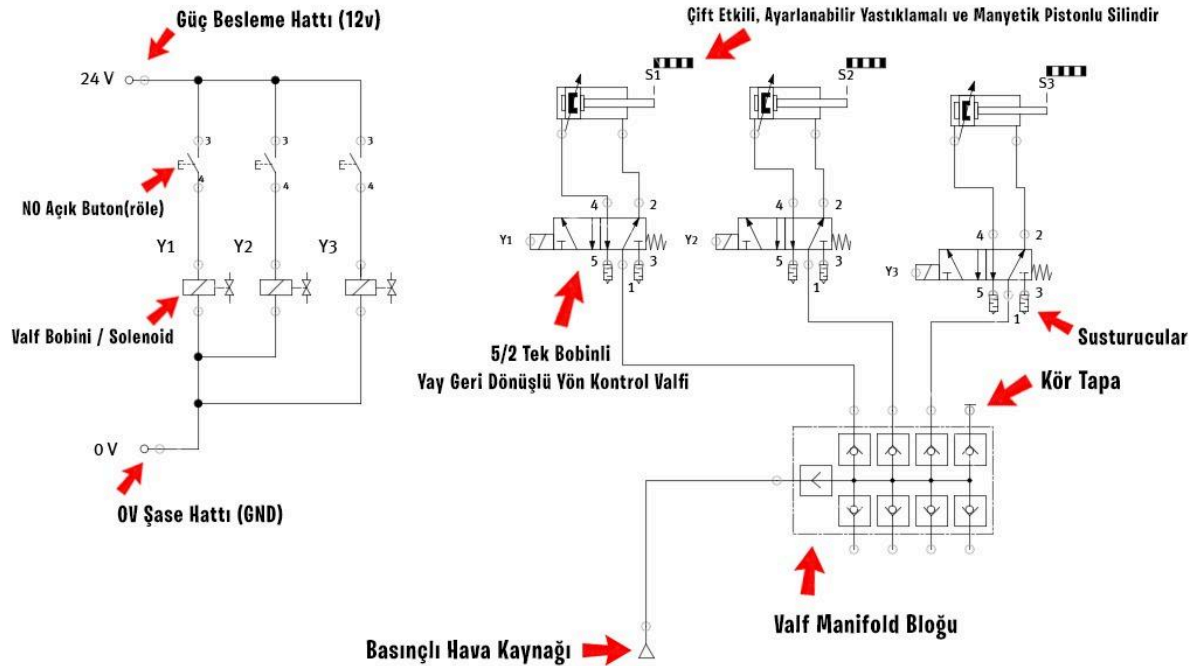
- **Frame Alignment:** Rigid vertical columns and base plate to ensure concentricity between the bottle neck and the decapping tool.
- **Leverage & Engagement:** The decapping head features an ergonomic mechanical hook/gripper geometry engineered to latch beneath standard bottle cap lips with uniform stress distribution.

2.2 Pneumatic Actuation & Control

The actuation system utilizes fluid power to deliver rapid, consistent strokes:

- **Pneumatic Cylinder:** Sized to provide sufficient linear thrust at standard working pressures (4–6 bar) to overcome cap retention resistance.
- **Valving & Control:** Directional control valves (manual/solenoid actuated) regulate the extension and retraction cycles of the piston.
- **Flow Control:** Integrated throttle valves prevent excessive shock loads on the glassware during downward/upward strokes.

Figure 1 : Pneumatic circuit diagram and actuator layout.



3. Fabrication and Assembly

The prototype fabrication followed a hybrid manufacturing workflow:

1. **Component Machining & Sourcing:** Structural plates, linkages, and mechanical guides were prepared, aligned, and toleranced to prevent binding during actuation.
2. **System Integration:** Pneumatic fittings, polyurethane tubing, and the main actuator were integrated into the mechanical chassis.
3. **Safety Considerations:** Adequate clearances and physical shielding were factored in to protect the operator during high-speed actuation.

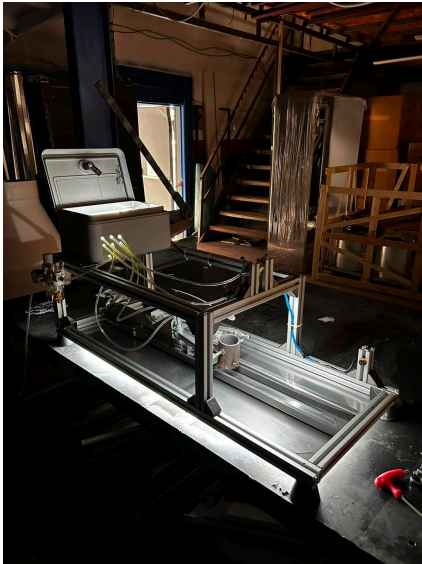


Figure 2: Fully assembled PBOS prototype

4. Experimental Testing & Results

The PBOS prototype underwent functional verification tests under varying operating pressures:

- **Cycle Time:** Cap removal was achieved in under 3 seconds per cycle.
- **Structural Integrity:** No permanent deformation was observed in the mechanical linkages or frame under repeated cyclic loads.
- **Consistency:** The system consistently decapped standard sample bottles without damaging the glass neck or causing pressure loss in the pneumatic line.

5. Conclusion & Future Work

The PBOS project demonstrates the successful application of pneumatic automation to everyday mechanical problems. The system provides a clean, safe, and repeatable opening cycle through optimized pneumatic force transmission.

Future iterations will explore:

- Automatic bottle detection using optical/proximity sensors.
- Microcontroller-based PLC automation for fully unattended continuous operation.
- Topology optimization of the mechanical frame to reduce total prototype weight.

References

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PBOS Project Documentation: <https://pbosproje.netlify.app/>